COVID-19 and Lung Disease Detection from Chest X-rays

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1. Problem and Objective

The task is to classify a chest X-ray into one of four classes: COVID-19, Lung opacity, Viral pneumonia or Normal. Fast, automated triage can support radiologists when scan volume is high. Because the images were pooled from many hospitals, a network may learn shortcuts (text markers, borders) instead of lung pathology, so Grad-CAM is used to check where the model looks. Success is judged by test accuracy and **macro F1**, which weights the small Viral pneumonia class equally, together with sensible Grad-CAM evidence.

2. Dataset and Pre-processing

The *COVID-19 Radiography Database* (Kaggle, tawsifurrahman) was used in full: 21,165 images in four classes (Table 1); the lung masks were not used. A two-step stratified split with a fixed seed (42) gave **15,291 train / 2,699 validation / 3,175 test** images (about 72/13/15). Images were resized to 224×224 and rescaled to [0,1]. Training-only augmentation: rotation ±15°, width/height shift 0.1, zoom 0.1 and horizontal flip. Class imbalance (Normal is about 7.6× larger than Viral pneumonia) was handled only by stratified splitting and macro-averaged metrics; no class weights or resampling were used.

Class	COVID-19	Lung opacity	Normal	Viral pneumonia	Total
Images (all)	3,616	6,012	10,192	1,345	21,165
Test set	542	902	1,529	202	3,175

Table 1. Class distribution and test-set size.

3. Methodology

An ImageNet-pretrained **MobileNetV2** was used as a **fully frozen feature extractor** (transfer learning, no backbone fine-tuning). The head is Global Average Pooling → Dense(128, ReLU) → Dropout(0.3) → Dense(4, softmax), giving 2.42 M parameters, of which only 164,484 are trainable. The model was trained with categorical cross-entropy and Adam for 10 epochs; no early stopping or learning-rate schedule was used, so the final-epoch weights were evaluated. Hyperparameters are in Table 2.

Backbone	Frozen	Loss	Optimiser	LR	Batch	Epochs	Early stop	Dropout
MobileNetV2 (ImageNet)	All layers	Categorical CE	Adam	1e-3 (default)	32	10	None	0.3

Table 2. Hyperparameters.

4. Results

Model	Test acc.	Macro F1	F1 COVID-19	F1 Lung opacity	F1 Normal	F1 Viral pneu.	Train time
MobileNetV2 (frozen)	0.87	0.87	0.85	0.81	0.90	0.92	≈36.4 min (10 epochs)

Table 3. Results on the **test set** (3,175 images). Per-class precision / recall: COVID-19 0.81/0.90, Lung opacity 0.93/0.71, Normal 0.85/0.95, Viral pneumonia 0.96/0.88.

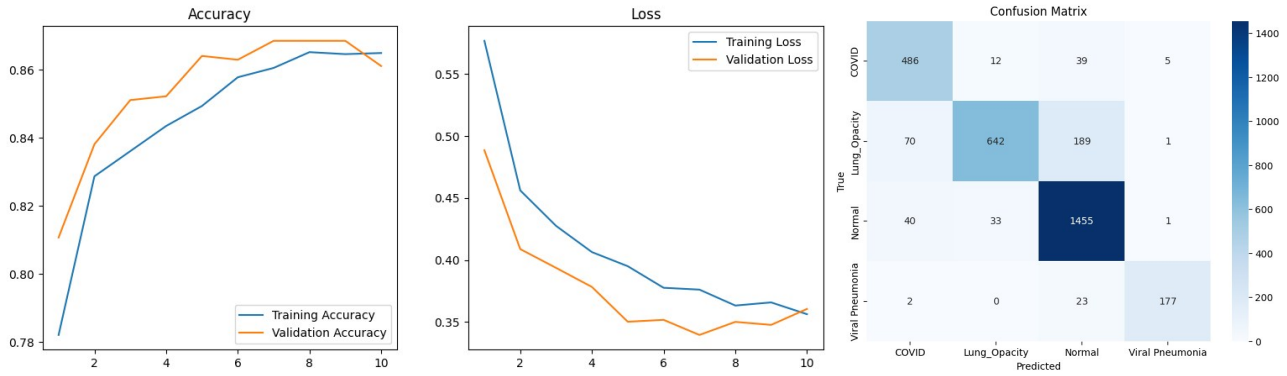


Fig. 1. Left: accuracy and loss curves (train vs validation, per epoch). Right: confusion matrix on the test set.

5. Analysis and Discussion

The dominant error is **Lung opacity → Normal** (189 of 902; recall only 0.71) followed by Lung opacity → COVID-19 (70); this also lowers COVID-19 precision to 0.81. Normal is confused with COVID-19 (40) and Lung opacity (33), and 23 of 202 Viral pneumonia images are called Normal. Mild opacities look similar to healthy lungs, and COVID-19 and Lung opacity share diffuse haziness. Training accuracy (0.86) and validation accuracy (0.86–0.87) stay close, so there is **no strong over-fitting**; the model is more likely **under-fitting** because the whole backbone is frozen, which explains the lower accuracy. Validation loss is lowest at epoch 7 (0.34) and rises slightly to 0.36 at epoch 10, a mild warning that training for longer without early stopping may not help. A total of 415 test images were misclassified; four are shown in Fig. 2.

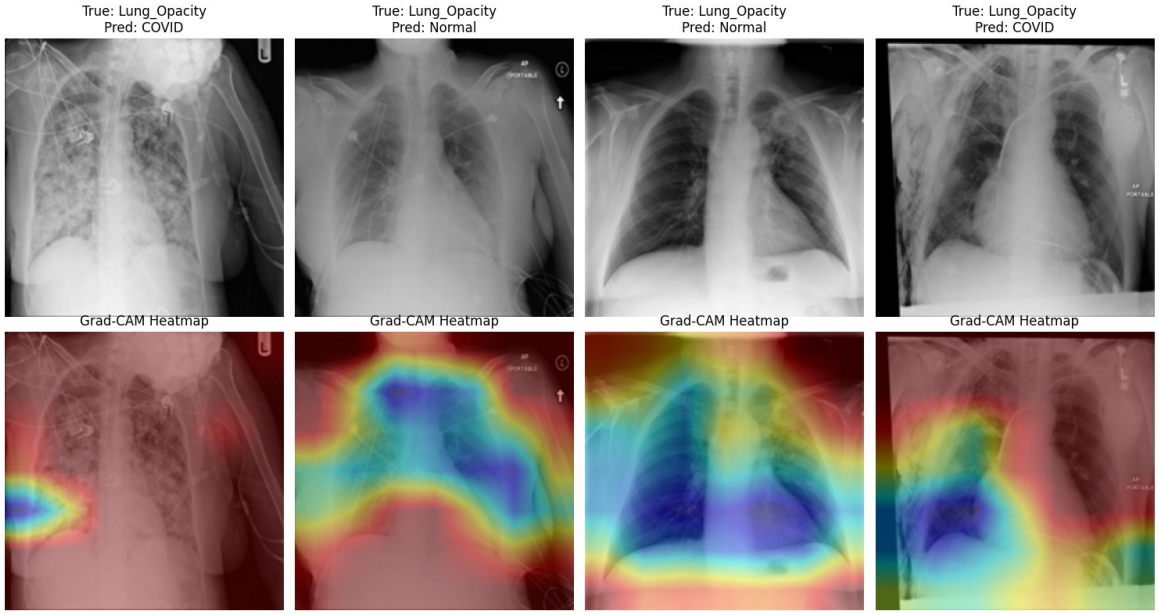


Fig. 2. Four misclassified test images (top row) and their Grad-CAM overlays (bottom row). All four are Lung opacity, predicted COVID-19, Normal, Normal and COVID-19 (left to right). In these overlays the colour channels are swapped, so **blue marks the strongest activation** and dark red the weakest.

Shortcut check. In the second to fourth images the strongest activation lies over the upper chest and lung fields, which is the expected region; in the first image the strongest region is a patch at the extreme left edge, outside the lungs, which could indicate reliance on an edge or an artefact. The images also contain text markers (“AP PORTABLE”, “L”), tubes and a tilted frame with black borders. Four examples cannot rule shortcut learning in or out, so the evidence is **inconclusive**. Likely reasons for the errors: faint or diffuse opacity (images 2 and 3, predicted Normal), and extensive hazy opacity that resembles COVID-19 plus overlying tubes and an off-centre or tilted view (images 1 and 4).

6. Real-world Relevance, Limitations and Ethics

The model could serve as a second-reader or triage aid where radiologists are scarce. It is not deployable today: accuracy of 0.87 and Lung-opacity recall of 0.71 would miss too many abnormal scans, it was validated only on one pooled public dataset with no external-hospital test, patients may appear in both train and test splits (no patient IDs), and the shortcut analysis is limited. Source bias (different scanners and demographics per class), patient privacy of medical images and the cost of false negatives mean it should assist, never replace, clinicians.

7. Conclusion and Future Work

A frozen MobileNetV2 reached **0.87 test accuracy and 0.87 macro F1** on four chest X-ray classes, with Lung opacity the hardest class. Next, I would fine-tune the upper backbone layers at a low learning rate with early stopping and class weights, and run Grad-CAM on many images per class (with border cropping) to test for shortcuts.

Dataset: M.E.H. Chowdhury et al., “Can AI help in screening viral and COVID-19 pneumonia?”, IEEE Access, 8, 2020; T. Rahman et al., Computers in Biology and Medicine, 132, 2021. Kaggle: COVID-19 Radiography Database.